Pegah Mohammadipour

Ph.D. Candidate in Mathematics

Department of Mathematics, The Pennsylvania State University
University Park, PA, USA

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Research Interests

Numerical analysis; quantum algorithms for open quantum systems; quantum simulation and error mitigation; mathematical reasoning and evaluation of large language models.

Education

Ph.D. in Mathematics Aug. 2022–Present
The Pennsylvania State University, University Park, PA
Advisor: Xiantao Li. Research focus: quantum algorithms for open quantum systems.

B.Sc. in Mathematics and Applications Sept. 2017–Sept. 2021
University of Tehran, Tehran, Iran
Undergraduate thesis: *The Geometric Concept of Parallel Transport along Curves with some Physical Interpretations*.

Minor in Computer Engineering Feb. 2019–Aug. 2021
University of Tehran, Tehran, Iran

Diploma in Mathematics and Physics Sept. 2010–July 2017
Aboureihan Educational Complex, Tehran, Iran

Research Experience

Research Intern June–Aug. 2026
Fujitsu Research America, Santa Clara, CA
Developed quantum algorithms for applications in finance.

Research Assistant Sept. 2021–May 2022
Institute for Research in Fundamental Sciences (IPM), Tehran, Iran
Studied functional inequalities for quantum correlations in networks.
Advisor: Salman Beigi.

Research Assistant July 2019–Sept. 2020
School of Electrical and Computer Engineering, University of Tehran
Worked on deep learning methods for image compression. Advisor: Farhad Pakdaman.

Awards & Honors

ICM Travel Grant, International Congress of Mathematicians 2026

Second Place, Penn State ICDS Symposium Poster Competition 2025

Honorable Mention, Mary Lister McCammon Award for Distinguished Undergraduate Teaching

Graduate Teaching Associate Status, recognition of commitment to undergraduate teaching

Top 0.5%, Iranian National University Entrance Examination, Mathematics and Physics 2017

Publications & Preprints

Journal articles

[1] **Reducing Circuit Depth in Lindblad Simulations via Step-Size Extrapolation**
Pegah Mohammadipour and X. Li.
Physical Review A **112**(6), 062206 (2025). doi:10.1103/dzpc-m1p4.

[2] **Direct Analysis of Zero-Noise Extrapolation: Polynomial Methods, Error Bounds, and Simultaneous Physical-Algorithmic Error Mitigation**
Pegah Mohammadipour and X. Li.
Quantum **9**, 1909 (2025). doi:10.22331/q-2025-11-14-1909.

Conference and workshop papers

[3] **CombiGraph-Vis: A Curated Multimodal Olympiad Benchmark for Discrete Mathematical Reasoning**
H. Mahdavi, P. Mahdavinia, A. Farhadi, **Pegah Mohammadipour**, S. Malek, M. Daliri, Pedram Mohammadipour, A. Hashemi, A. Khasahmadi, and V. Honavar.
NeurIPS 2025 Workshop MATH-AI. arXiv:2510.27094.

[4] **RefGrader: Automated Grading of Mathematical Competition Proofs using Agentic Workflows**
H. Mahdavi, P. Mahdavinia, S. Malek, **Pegah Mohammadipour**, A. Hashemi, M. Daliri, A. Farhadi, A. Khasahmadi, N. Mireshghallah, and V. Honavar.
NeurIPS 2025 Workshop MATH-AI. arXiv:2510.09021.

[5] **Brains vs. Bytes: Evaluating LLM Proficiency in Olympiad Mathematics**
H. Mahdavi, A. Hashemi, M. Daliri, **Pegah Mohammadipour**, A. Farhadi, S. Malek, Y. Yazdanifard, A. Khasahmadi, and V. Honavar.
Conference on Language Modeling (COLM), 2025. arXiv:2504.01995.

Preprint

[6] **MLMC-qDRIFT: Multilevel Variance Reduction for Randomized Quantum Hamiltonian Simulation**
Pegah Mohammadipour and X. Li.
2026. arXiv:2604.26865.

Talks

From the Heat Equation to Lindblad Dynamics Oct. 2026
Invited talk. Penn State Partial Differential Equation Graduate Student Seminar.
The Pennsylvania State University, University Park, PA.

Reducing Circuit Depth in Lindblad Simulation via Step-Size Extrapolation Mar. 2026
Contributed talk. APS Global Physics Summit.
Denver, CO.

Polynomial Extrapolation for Lindblad Simulation Jan. 2026
Lightning poster talk. New Frontiers in Quantum Algorithms for Open Quantum Systems.
Institute for Pure and Applied Mathematics (IPAM), UCLA, Los Angeles, CA.

Polynomial Extrapolation for Error Mitigation in Quantum Simulation Nov. 2025
Contributed talk. SIAM New York–New Jersey–Pennsylvania Section Annual Meeting.
The Pennsylvania State University, University Park, PA.

Poster Presentations

Polynomial Extrapolation for Physical Noise and Step-Size Error Mitigation in Lindblad Simulation Aug. 2026
Quantum Computational Sensing Workshop, Stanford Q-FARM.
Stanford University, Stanford, CA.

Polynomial Extrapolation for Lindblad Simulation Jan. 2026
New Frontiers in Quantum Algorithms for Open Quantum Systems.
IPAM, UCLA, Los Angeles, CA.

Polynomial Extrapolation for Physical Noise and Step-Size Error Mitigation in Lindblad Simulation Nov. 2025
Russell E. Marker Lectures in Mathematics, Poster Session.
The Pennsylvania State University, University Park, PA.

Polynomial Extrapolation for Physical Noise and Step-Size Error Mitigation in Lindblad Simulation Oct. 2025
Institute for Computational and Data Sciences (ICDS) Symposium.
The Pennsylvania State University, University Park, PA.

Polynomial Extrapolation for Physical Noise and Step-Size Error Mitigation in Lindblad Simulation Aug. 2025
QSim 2025 Conference.
IBM, New York, NY.

Teaching Experience

Graduate Teaching Associate Aug. 2022–Present
Department of Mathematics, The Pennsylvania State University
MATH 231: Multivariable Calculus.

Teaching Assistant
Department of Mathematics, The Pennsylvania State University
MATH 455: Introduction to Numerical Analysis I Aug.–Dec. 2026
MATH 467: Factorization and Primality Testing Aug.–Dec. 2022

Teaching Assistant Sept. 2019–Jan. 2022
School of Mathematics, Statistics and Computer Science, University of Tehran
Differential Equations Fall 2021
Data Structures Fall 2020
Calculus II Spring 2020
Introduction to Computer Programming Fall 2019

English Teacher Mar.–Oct. 2018
Safir Language Academy, Tehran, Iran

Professional Service

Reviewer, NeurIPS MATH-AI Workshop 2026
Reviewer, *Journal of Computational Physics* 2026
Reviewer, 26th Asian Quantum Information Science Conference (AQIS) 2026

Leadership & Volunteering

Organizer, Quantum Computation Reading Group Fall 2025
The Pennsylvania State University, University Park, PA

Poster Judge, ECoS Undergraduate Research Exhibition Fall 2024
The Pennsylvania State University, University Park, PA

Editor-in-Chief, *Jonge Riazi* Student Mathematics Magazine Jan. 2020–Sept. 2021
School of Mathematics, Statistics and Computer Science, University of Tehran
Revived the magazine after a 24-year hiatus; recruited and mentored a student editorial team to develop, edit, and prepare articles for publication.

Elected Committee Member, Mathematics Student Association Sept. 2018–Sept. 2020
University of Tehran, Tehran, Iran
Organized *Math Suppers*, a weekly series of mathematics talks; coordinated student study groups and workshops.

Conferences & Summer Schools

July 2026 **International Congress of Mathematicians (ICM)**. Philadelphia Convention Center, Philadelphia, PA, and online; July 23–30.

Mar. 2026 **APS Global Physics Summit**. Colorado Convention Center, Denver, CO, and online; March 15–20.

Jan. 2026 **New Frontiers in Quantum Algorithms for Open Quantum Systems**. IPAM, UCLA, Los Angeles, CA; January 12–16.

Aug. 2025 **QSim Conference**. IBM, New York, NY.

Aug. 2024 **QSim Conference and Summer School**. University of Rhode Island, Kingston, RI.

July 2024 **Graduate Summer School in Stochastic Quantization**. SLMath, Berkeley, CA.

July 2023 **Graduate Summer School in Concentration Inequalities and Localization Techniques in High Dimensional Probability and Geometry**. SLMath, Berkeley, CA.

2021 **Undergraduate Summer School on Experimental QIP (USEQIP)**. University of Waterloo, Canada; remote, May–July.

July 2021 **Qiskit Global Summer School: Quantum Machine Learning**. IBM Quantum; remote.

July 2020 **Qiskit Global Summer School: Introduction to Quantum Computing and Quantum Hardware**. IBM Quantum; remote.

Aug. 2019 **Game Theory Summer School**. IPM, Tehran, Iran.

Aug. 2018 **Deep Learning Summer School (UTDLSS)**. School of Electrical and Computer Engineering, University of Tehran.

Professional Memberships

Society for Industrial and Applied Mathematics (SIAM), Graduate Student Member Sept. 2026–Present

Institute of Electrical and Electronics Engineers (IEEE), Graduate Student Member Sept. 2026–Present

American Physical Society (APS), Graduate Student Member Nov. 2025–Present

Skills

Programming: Python, MATLAB, C, C++, R, Octave, HTML/CSS.
Software & tools: Qiskit, LaTeX, Jupyter Notebooks; Ubuntu and Windows.
Languages: Persian (native), English (fluent), Spanish (intermediate).